

1.3 Charges in Uniform Fields

Date: _____

1. An electron enters a region of uniform field $E = 500 \text{ N/C}$ with speed $2.0 \times 10^6 \text{ m/s}$ perpendicular to the field. How far has it been deflected sideways after travelling 4.0 cm along its original direction? Ignore gravity.

2. An electric dipole with dipole moment $p = 6.0 \times 10^{-30}$ C·m sits in a uniform field $E = 1.0 \times 10^5$ N/C, making a 30° angle with the field. Find the torque on the dipole and its potential energy.